Claws, boutons, calyx geometry and the outside-ROI version

> **QUESTION**
>
> How are PN boutons and KC claws found, how are APL synapses placed on them, where on the KC do the other APL synapses sit, and which synapses does the corrected calyx add?

INPUTS

- PN → KC synapses in the corrected calyx (positions of T-bars and PSDs)
- Every synapse site of every KC near the calyx
- KC skeletons: hemibrain v1.2 (release archive) and male CNS v1.0 (public SWC files)
- Prat's claw counts per hemibrain KC (uploaded `KC_Claw_Counts_Hemibrain` file)
- The APL synapse tables of Report 01

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/` (claws, the APL tables with the new claw columns, KC ↔ APL on claws)
- `claws_geodesic/` (the claw comparisons and the 40 example KCs)
- `kc_skeletons/claws_v2m2/` (KC skeletons with APL synapses)
- `<dataset>_PN_boutons_derived.csv.gz`
- `with_outside_ROI_synapses/` (the corrected calyx)

SCRIPTS

- `scripts/claws_geodesic/extract_kc_sites.py`, `segment_claws.py`, `claws_v2.py` (claws)
- `scripts/claws_geodesic/compare_claws.py`, `compare_three.py` (comparisons)
- `scripts/claws_geodesic/export_claws_v2.py`, `export_claws_v2_tables.py` (tables for the other scripts)
- `scripts/kc_skeleton_apl_synapses_v2.py` (KC skeletons)
- `scripts/run.py`, `scripts/core.py` (boutons)
- `scripts/offroi/assign.py` (corrected calyx)
- `scripts/reports/r02_prep_v2.py`, `scripts/reports/r02_claws_boutons.py` (this report)

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5 Where the off-claw APL synapses sit on the KC

6 Drawing the calyx: the posterior view

7 The corrected calyx: synapses just outside the official ROIs

None of the four connectomes publishes which synapses form a PN bouton or a KC claw, so both are derived here. They matter because odor information enters the calyx bouton by bouton: each bouton belongs to one PN and so to one glomerulus, and each claw takes its input from one PN. Reports 03, 04, 05 and 10–12 use claws and boutons to give APL synapses a glomerulus.

What changed in this version (2026-11-04)

All tables are the **corrected calyx** (synapses just outside the official ROIs added; last section).

New claws, found along each KC's skeleton with the rule Yang gave on 2026-11-04 and the variant Yang chose (a PN needs at least 2 PSDs to get its own claw). The old claws were 2 µm straight-line clusters of PN → KC PSDs.

The new claws exist for **hemibrain and male CNS**, the two datasets with KC skeletons. FlyWire and BANC keep the old 2 µm claws where they appear (Report 04); the model (Reports 05–12) uses only hemibrain and male CNS.

New: Figure 1 (3 random PNs and 10 random KCs with their skeletons), the comparison with Prat's claw counts, APL ↔ KC figures in the calyx, and KC skeletons showing where the off-claw APL synapses sit.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
$d_{\text{geo}}(s, s')$	Distance between sites s and s' of one KC measured along its skeleton	µm
d_{cut}	Cut for the claw clusters and for joining a claw	8 µm, along the skeleton
m	Fewest PSDs a PN needs in a cluster to get its own claw	2 (Yang's choice)
d_{link}	Linking distance for boutons: two T-bars of one PN closer than this are in the same bouton	2 µm, straight line
$\mathbf{x}_s$	Position of site s (T-bar or PSD)	nm, native coordinates
$\pi(c)$	The PN of claw c (each new claw has exactly one)	neuron ID
$\delta(e)$	Distance along the KC from the KC-side site of APL synapse e to the nearest PSD of a claw's own PN	µm, column `claw_dist_nm`
$(R,\ D,\ A)$	Posterior-view axes: fly's right, dorsal, anterior–posterior	µm

1 Boutons and claws

1.1 Boutons: straight-line clusters of a PN's T-bars

Take every T-bar of one antennal-lobe PN that has a partner in the calyx (one point per T-bar, however many PSDs it faces). Link any two T-bars closer than $d_{\text{link}} = 2$ µm. Each connected group is one bouton. This rule is unchanged.

$$s \sim s' \iff \exists\, s = s_0, s_1, \ldots, s_m = s' \ (\text{T-bars of one PN}) : \ \|\mathbf{x}_{s_l} - \mathbf{x}_{s_{l-1}}\| < d_{\text{link}} \ \ \forall\, l \tag{1}$$

Two T-bars are in the same bouton when such a chain links them, so a bouton can be longer than 2 µm.

1.2 Claws: clusters along the KC skeleton, split by PN

The new claws follow the rule Yang gave on 2026-11-04. They use every synapse site of the KC in the calyx: PSDs from any partner and the KC's own T-bars. A site is in the calyx if it is labelled CA or lies within 3 µm of a corrected-calyx PN → KC PSD. Each site is placed on the nearest node of the KC's skeleton, and distances are measured along the skeleton.

1. **Cluster.** Link any two sites of the KC whose distance along the skeleton is at most $d_{\text{cut}} = 8$ µm. The connected groups are the clusters. This is the same as building the minimum spanning tree of the along-skeleton distances and cutting every edge longer than 8 µm.
2. **Split by PN.** In each cluster, every PN that makes at least $m = 2$ PSDs there gets its own claw, made of its PSDs.
3. **Join.** Every other site in the cluster joins the claw whose own PN's PSDs are nearest along the skeleton, if they are within 8 µm. This covers PSDs from APL and other partners, the KC's T-bars, and PSDs of a PN with a single PSD there. A site further than 8 µm from every claw is not on a claw.

$$\text{claw}(e) = \arg\min_{c} \min_{s \in \text{PSDs of } \pi(c) \text{ in } c} d_{\text{geo}}(e, s), \qquad e \text{ is on a claw} \iff \delta(e) = \min_{c} \min_{s} d_{\text{geo}}(e, s) \le 8\ \mu\text{m} \quad (2)$$

Rule 3 for a site e that is not a PSD of a claw's own PN.

Why $m = 2$. With $m = 1$, the rule as first written, a single stray PSD from a PN makes a claw of its own: 13% of hemibrain claws had one PSD, and hemibrain KCs had 7.3 claws against 6.7 in Prat's counts. Requiring at least 2 PSDs gave the closest match to Prat's counts of every version tried, and Yang chose it on 2026-11-04 (`claws_geodesic/README.md` has m = 1, 2, 3 and 5).

Skeletons. Hemibrain: the v1.2 release archive. Male CNS: the public v1.0 SWC files (coarse, about 0.7 µm between nodes). Pieces of a skeleton that are not connected are joined by their shortest gaps first. A site sits a median of 0.19 µm from its skeleton node in hemibrain and 0.63 µm in male CNS (90%: 0.42 and 1.18 µm).

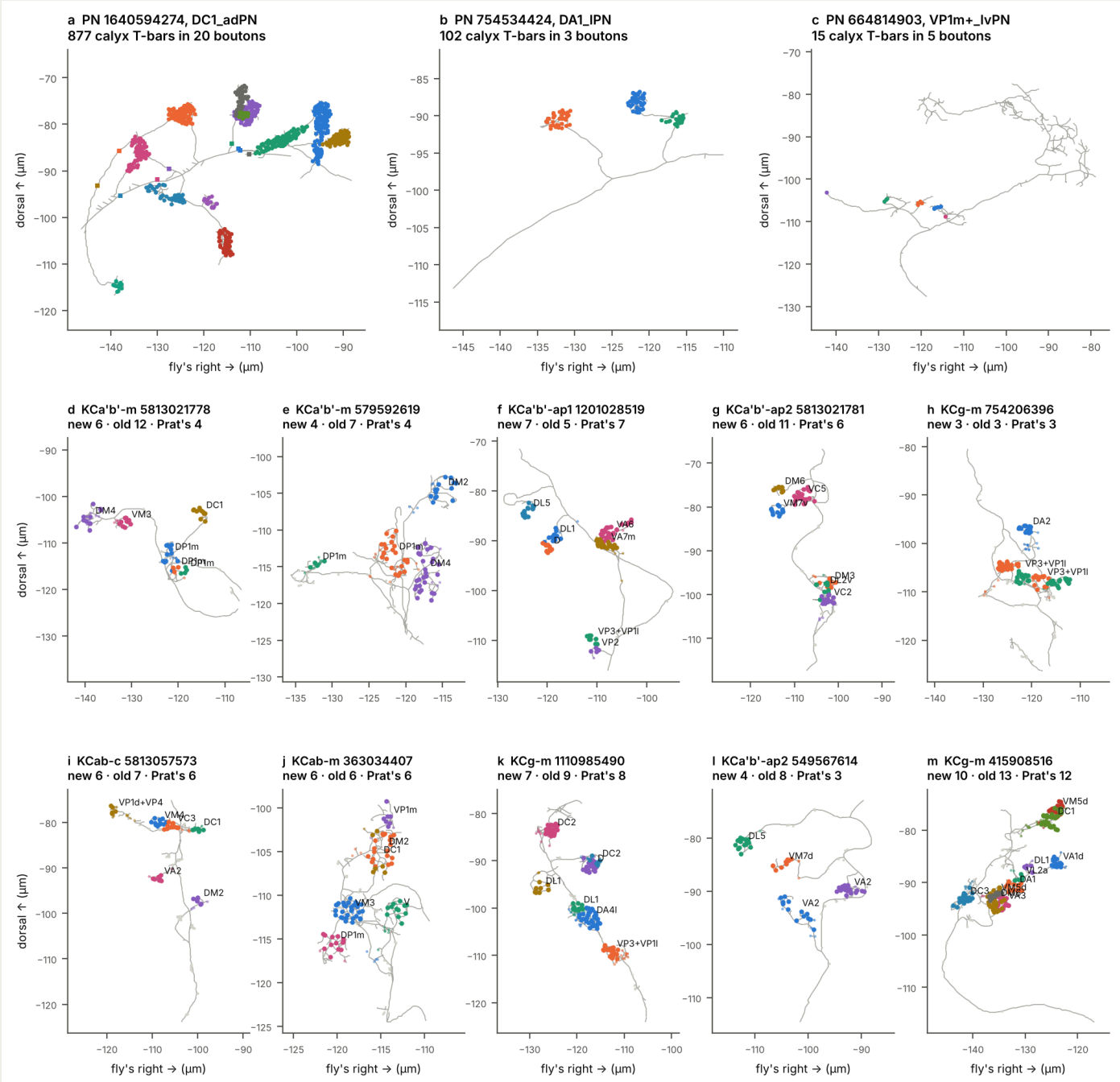


Figure 1. Three random PNs and ten random KCs with their skeletons. Hemibrain right calyx, posterior view (fly's right to the right, dorsal up, µm). Cells drawn at random among those with a skeleton (seed 1103): **a–c**, three uniglomerular PNs; **d–m**, ten KCs. Grey lines: the part of each skeleton inside the calyx (nodes within 3–4 µm of calyx synapse sites). **a–c**, every calyx T-bar of the PN, colored by its bouton (colors only separate boutons). **d–m**, the KC's calyx synapse sites. Large dots are PN → KC PSDs, small dots the other sites (PSDs from other partners, the KC's T-bars), both colored by their new claw; light grey = not on a claw. The label next to each claw is its PN's glomerulus. Each title gives the claw count from the new rule, the old 2 µm rule and Prat's segmentation (corrected count).

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2 New claws against the old ones and Prat's segmentation

Prat's segmentation is the uploaded hemibrain file of claw counts per KC (`n_claws` and, after correction, `n_claws_corrected`; the corrected count is used). It has counts only, no claw positions, so it can be compared per KC but not drawn. 1,800 hemibrain KCs are in all three. 83 of Prat's KCs are left out because they have no PN input in our corrected calyx (73 are labelled "KC part due to gap").

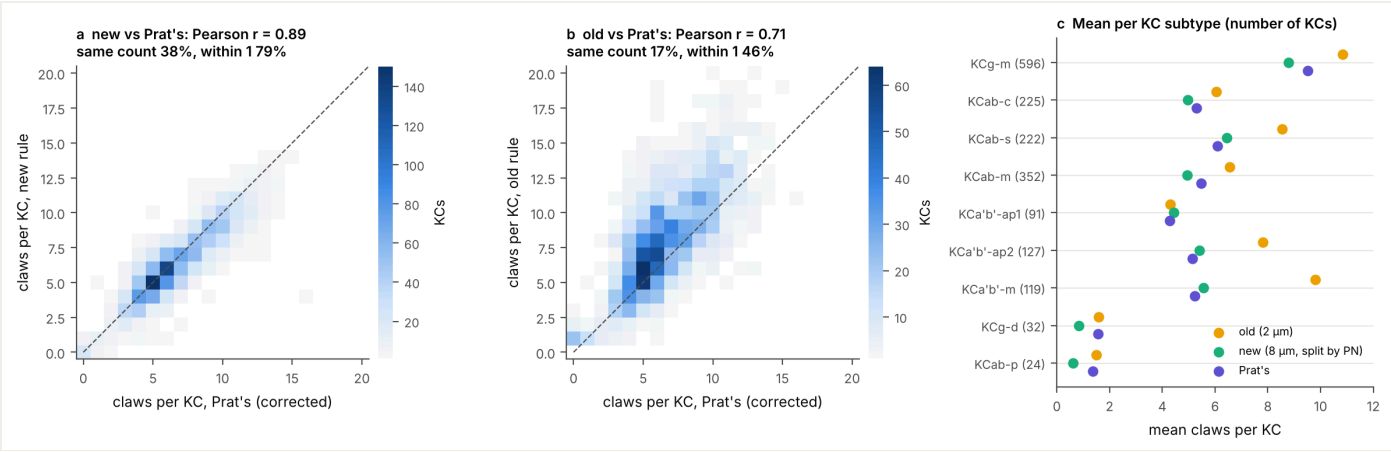


Figure 2. Claws per hemibrain KC: new, old and Prat's. **a**, **b**, Each cell counts the KCs with a given pair of claw counts (color, white = none); the dashed line is equality. **a**, new rule against Prat's; **b**, old 2 µm rule against Prat's. **c**, mean claws per KC for each hemibrain KC subtype under the three segmentations (yellow old, green new, violet Prat's). The full comparison, with the uncorrected counts and difference histograms, is `claws_geodesic/Yang_rule_2026-11-04/chosen_min2PSD/three_way_claw_comparison.png`.

Table 2. How well the claw counts agree. Per-KC agreement over 1,800 hemibrain KCs. Means: old 8.2, new 6.3, Prat's 6.7 (uncorrected 6.1) claws per KC.

pair	Pearson r	mean difference	same count	within 1	within 2
new vs Prat	0.89	−0.33	38%	79%	95%
old vs Prat	0.71	+1.54	17%	46%	67%
new vs old	0.79	−1.86	17%	46%	66%
new vs Prat (uncorrected)	0.81	+0.21	30%	71%	90%
old vs Prat (uncorrected)	0.66	+2.08	13%	39%	58%

The new rule matches Prat's counts much better than the old one (Pearson r 0.89 against 0.71; 79% of KCs within one claw, against 46%). The old rule overcounted most where a claw is long or branched (α′β′-m and γ-m KCs): 2 µm in a straight line cuts such a claw into pieces, while 8 µm along the skeleton keeps it whole. The new rule gives slightly fewer claws than Prat's on average.

Examples. `claws_geodesic/Yang_rule_2026-11-04/chosen_min2PSD/claw_disparity_KCs.pdf` (and a rotatable 3D view, `claw_disparity_KCs_3D.html`) shows the 40 KCs where the old and new claws differ most, old claws on the left and new on the right, Prat's count in each title. There, PN → KC PSDs are colored and shaped by the glomerulus of their presynaptic PN, the same in both panels, so the presynaptic partners can be seen not to change between segmentations; only their grouping into claws does.

3 How many claws and boutons, and what a claw holds

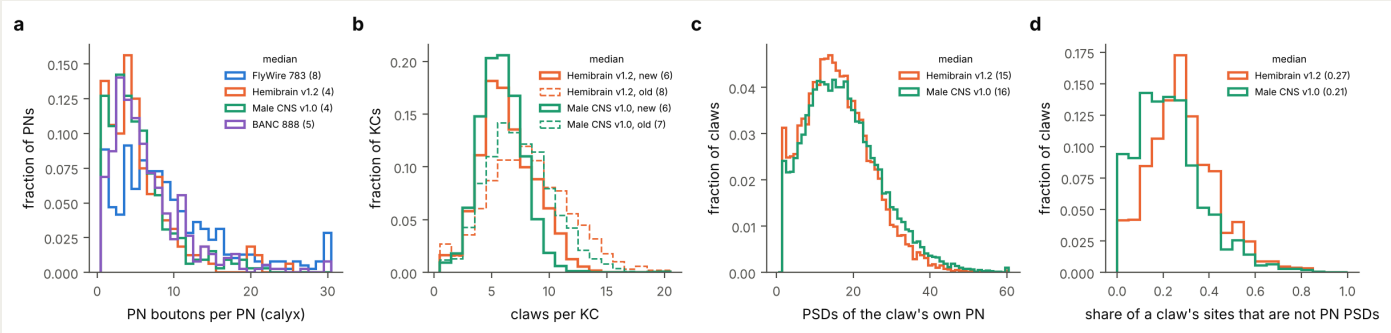


Figure 3. Boutons per PN, claws per KC, and what a claw holds. Line color = dataset; the number in brackets is the median. **a**, boutons per antennal-lobe PN in the calyx (all four datasets; FlyWire's table gives each connection its own presynaptic point, which likely explains its higher count). **b**, claws per KC with the new rule (solid) and the old 2 µm rule (dashed), KCs with at least one claw. **c**, PSDs from the claw's own PN. **d**, the share of a claw's sites that are not PN → KC PSDs (APL and other partners, the KC's T-bars).

Hemibrain KCs have a median of 6 new claws (8 with the old rule) and male CNS KCs 6 (7). Each new claw has one PN by construction; PSDs of a PN with a single PSD in the cluster can still join it as "other sites", so 98% (hemibrain) and 99% (male CNS) of claws get at least 90% of their PN PSDs from their own PN. 59% of hemibrain claws share their 8 µm cluster with another PN's claw, which is where the split by PN matters.

Table 3. Which sites are on claws. Share of each kind of KC calyx site that ends up on a claw.

sites	hemibrain	male CNS
all calyx sites	84%	88%
PN→KC PSDs	100%	100%
other PSDs (non-PN partner)	61%	62%
KC T-bars	54%	61%

4 Placing APL synapses on claws and boutons

KC partners. An APL ↔ KC synapse in the calyx is a site of the KC: a PSD when APL is presynaptic, a T-bar of the KC when the KC is. That site already has its claw from the rule above (step 3), so the synapse is on a claw exactly when its KC-side site is. Its distance $\delta(e)$ along the KC to the nearest PSD of a claw's own PN is stored as `claw_dist_nm`.

98.2% of hemibrain and 92.2% of male CNS KC ↔ APL calyx synapses are claw-segmentation sites. The rest, almost all in the back of the male CNS calyx, lie in the corrected calyx but more than 3 µm from any PN → KC PSD; they got the same rule straight on the skeleton, and 9% of them landed on a claw.

PN partners. APL ↔ PN synapses are placed on boutons as before: on the bouton of the nearest T-bar of that same PN, if within 2 µm.

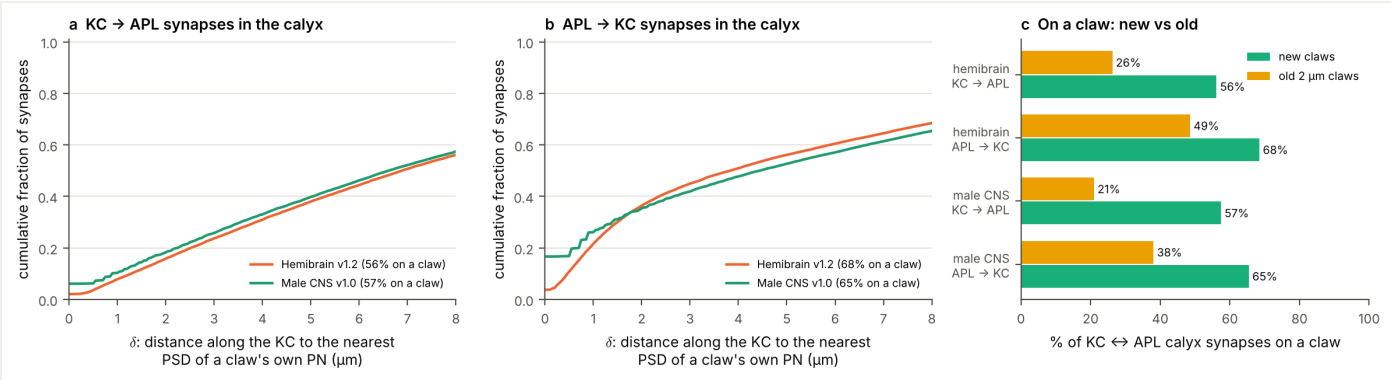


Figure 4. How far KC ↔ APL synapses lie from claws, and how many are on one. **a**, **b**, Cumulative distributions over all KC ↔ APL synapses in the corrected calyx, one line per dataset: the fraction whose KC-side site lies within the distance on the x-axis, measured along the KC skeleton, of a PSD of one of the KC's claws' own PNs. The axis ends at the 8 µm cut: distances were computed only up to 8 µm, so each curve ends at the share of synapses on a claw, and the rest are further than 8 µm along the KC (or in a cluster with no claw). **c**, Share of these synapses on a claw with the new rule (green) and with the old 2 µm rule (yellow).

With the new claws, most KC ↔ APL calyx synapses are on a claw: Hemibrain v1.2 56% of KC → APL and 68% of APL → KC, Male CNS v1.0 57% of KC → APL and 65% of APL → KC. The old 2 µm rule put only 21–49% on claws, because it measured straight-line distance to the PN PSDs and APL synapses often sit a few micrometres along the claw from them.

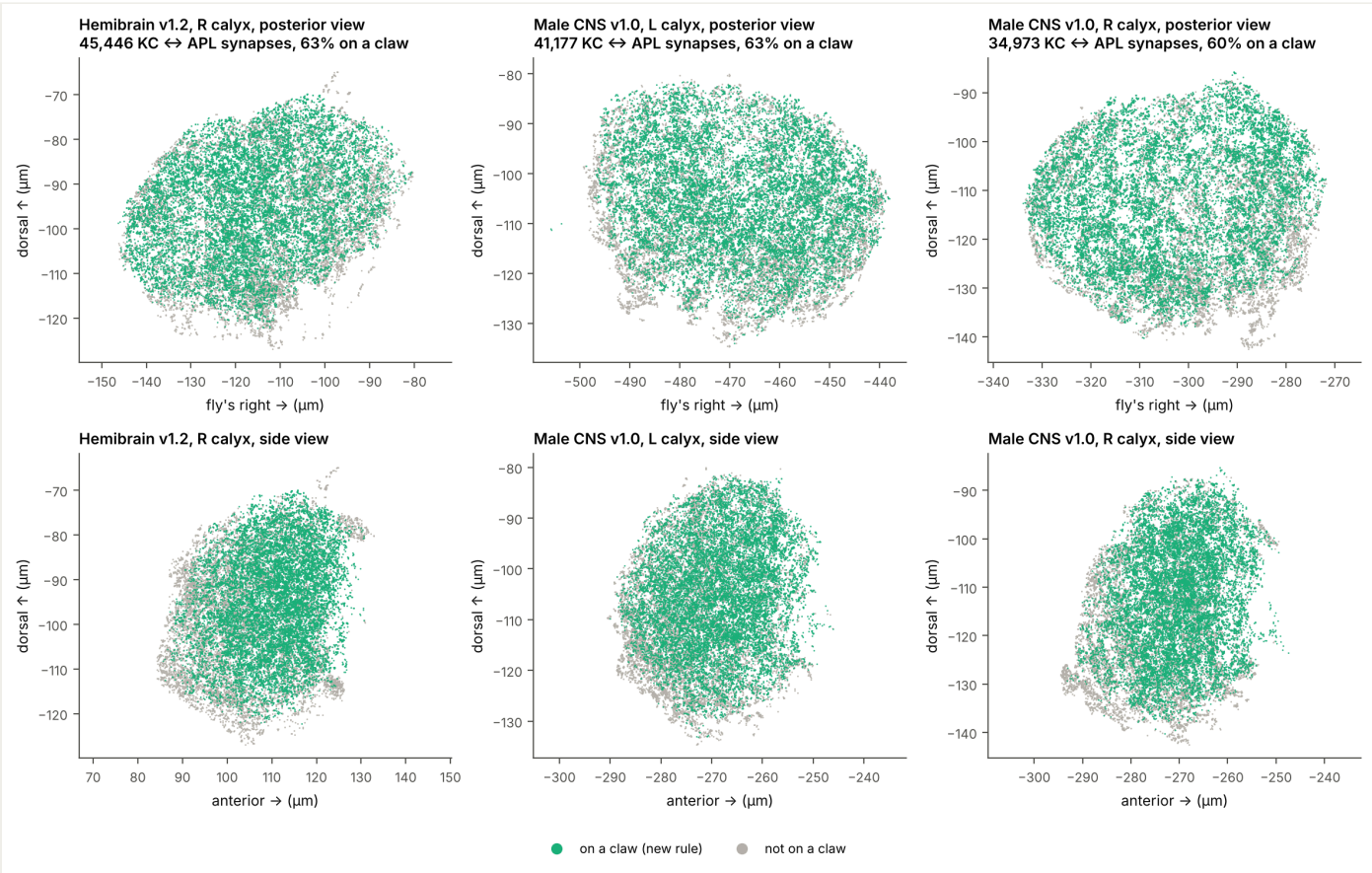


Figure 5. APL ↔ KC synapses in the calyx, on and off claws. Every KC ↔ APL synapse (both directions) in the three corrected calyces used by the model, drawn at its KC-side site. Green = on a claw, grey = not. Top: posterior view (axes of Table 4, µm). Bottom: side view, anterior to the right (sign set so that the lobes lie anterior of the calyx), dorsal up. At most 15,000 points of each color are drawn per panel (random subsample).

5 Where the off-claw APL synapses sit on the KC

For male CNS KCs of the right calyx, every calyx KC ↔ APL synapse was placed on the KC's skeleton and classified. The skeleton is rooted at its most anterior node (a lobe tip), so "downstream" means away from the axon. For the synapse's node, count the claws whose PSDs lie at or downstream of it:

- **on a claw**: the synapse is on a claw (rule above);
- **dendritic trunk**: not on a claw, upstream of two or more claws (a larger branch, before it splits into claws);
- **claw stalk**: upstream of exactly one claw (the branch leading to one claw, more than 8 µm from its PSDs);
- **branch without claw**: no claw downstream (a side branch, or the soma neurite).

Table 4. Where calyx KC ↔ APL synapses sit on the KC. All 1,901 right-calyx male CNS KCs with a skeleton, 34,562 calyx KC ↔ APL synapses. Off-claw synapses lie a median of 13 µm along the KC from the nearest PN → KC PSD (90% within 25 µm).

location	all	KC → APL	APL → KC
on claw	61%	57%	65%
dendritic trunk	16%	20%	13%
claw stalk	10%	11%	10%
branch without claw	12%	12%	13%

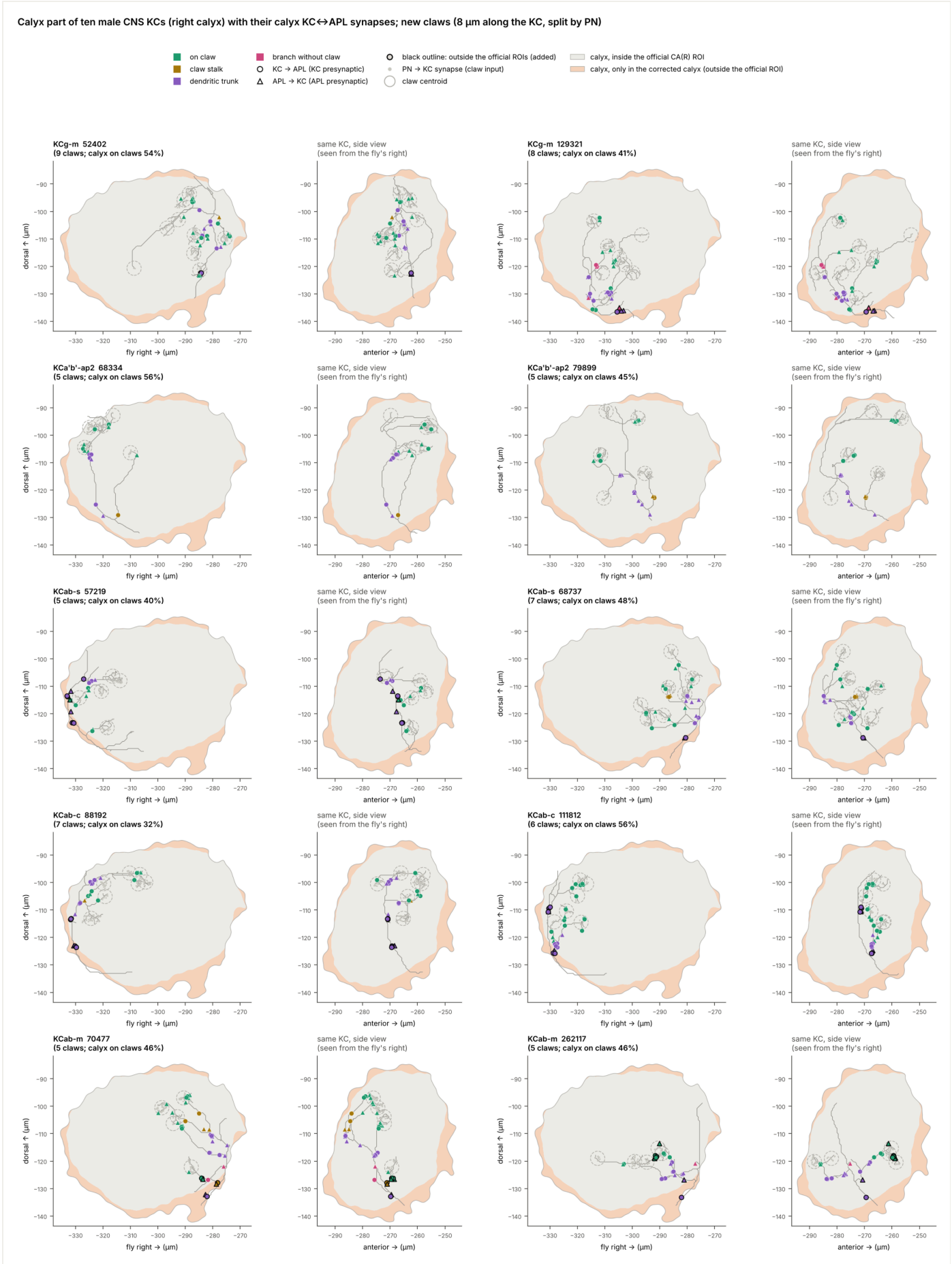


Figure 6. Ten KCs with off-claw APL synapses on their larger branches. The ten right-calyx male CNS KCs with the most calyx KC ↔ APL synapses on the dendritic trunk (at most two per KC type), so the panels show the case Yang asked about: APL synapses on the larger branches before they split into claws. For each KC, the posterior view (left) and the side view seen from the fly's right (right) of its calyx part. Grey lines: skeleton; small grey dots: PN → KC PSDs; dashed circles:

claw centroids. KC ↔ APL synapses: circles KC → APL, triangles APL → KC, colored by location (green on a claw, purple dendritic trunk, brown claw stalk, pink branch without claw); black outline = outside the official ROIs. Background: the corrected right calyx (grey inside the official ROI, orange only in the corrected calyx). Whole KCs and a rotatable 3D view are in `kc_skeletons/claws_v2m2/`.

Across all right-calyx KCs, 61% of calyx KC ↔ APL synapses are on claws. Of the rest, the largest group sits on the dendritic trunk (16% of all), then on branches without claws (12%) and on claw stalks (10%). KC → APL synapses are more often off-claw than APL → KC synapses (43% against 35%), and they sit on the trunk more often (20% against 13%).

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6 Drawing the calyx: the posterior view

Every 2D map in these reports shows the calyx as seen from behind the fly (posterior view), with the fly's right to the right and dorsal up, so that all datasets look alike. Each dataset has its own axes, so the view is built from its native coordinates (Table 4), checked against mushroom-body landmarks (the lobes are anterior and ventral of the calyx, the vertical lobe is dorsal).

Table 5. Native axes used for the posterior view. Coordinates are in nm in the tables and µm in the maps. The 3D maps of Report 03 use the same right and dorsal axes, with anterior signed so that the peduncle lies anterior of the calyx.

Dataset	Right (x of the map)	Dorsal (y of the map)	Anterior–posterior (depth)	Note
FlyWire 783	x	$-y$	z	FlyWire coordinates are mirror-imaged relative to its side labels, so sides come from the labels
Hemibrain v1.2	$-x$	$-z$	y	the hemibrain y axis is anterior–posterior
Male CNS v1.0	$-x$	$-y$	z	
BANC 888	$-x$	z	y	

The calyx outlines still differ between datasets in these maps, mainly because each brain sits at a slightly different angle. The calyx is a flattened cup; its thinnest axis is tilted from the anterior–posterior axis by 25° (Hemibrain v1.2 R), 18° (Male CNS v1.0 L), 25° (Male CNS v1.0 R), and by 35–41° in BANC, so BANC's calyx is seen more obliquely and looks flatter. `odor_weighted/calyx_shape_by_dataset.png` shows every calyx from three sides.

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7 The corrected calyx: synapses just outside the official ROIs

Each connectome defines its mushroom body regions with its own ROI meshes, and these do not always cover the whole structure. In the male CNS the calyx ROIs leave out the posterior part of each calyx; in FlyWire and BANC the meshes of nearby neuropils (CRE, SIP, SMP) take in parts of the lobes. `with_outside_ROI_synapses/` holds the tables with these synapses added back, and since 2026-11-03 it is the main version of every regenerated report (the official-ROI tables are kept unchanged). The rule, in order:

1. **Anchors.** Calyx: the PN → KC synapses inside the official calyx ROI, extended through PN → KC synapses outside every MB ROI that connect to them in steps of ≤ 3 µm (claws the ROI cut off). Peduncle and lobes: the APL ↔ KC synapses inside the official peduncle and lobe ROIs.

2. **Growing through KC synapses.** APL ↔ KC synapses outside every MB ROI join the first anchor that reaches them in steps of ≤ 3 µm, and take that anchor's region. KC dendrites and axons exist only in the mushroom body, so these chains stay inside it.
3. **Joining.** Any other synapse outside every MB ROI joins when its postsynaptic site lies within 3 µm of an anchor point, with the region of the nearest one (the calyx wins ties).
4. **Accessory calyces stay out.** No chain runs through synapses with γd or αβp KCs, whose dendrites form the accessory calyces.
5. **PN synapses** next to the calyx join only if they are PN → KC or their T-bar belongs to a calyx bouton.

Every added row is flagged in `mb_region_source`; keeping the rows marked `official ROI` gives the official tables back exactly.

Table 6. Synapses added in the corrected version. Added synapses as a count and as a share of the official-ROI count, from `with_outside_ROI_synapses/added_synapses_summary.csv`. The male CNS calyx gains the most (+31% of APL synapses).

Dataset	APL synapses, calyx	peduncle	lobes	KC ↔ APL, calyx	PN → KC, calyx
FlyWire 783	+3,055 (+3.6%)	+12,498 (+23.9%)	+34,685 (+24.6%)	+4.0%	+0.8%
Hemibrain v1.2	+1,607 (+2.6%)	+1,411 (+5.5%)	+791 (+0.5%)	+2.2%	+2.8%
Male CNS v1.0	+26,990 (+30.9%)	+4,872 (+11.9%)	+19,976 (+6.4%)	+32.0%	+14.0%
BANC 888	+753 (+2.0%)	+2,131 (+17.3%)	+9,771 (+56.5%)	+2.2%	+0.7%

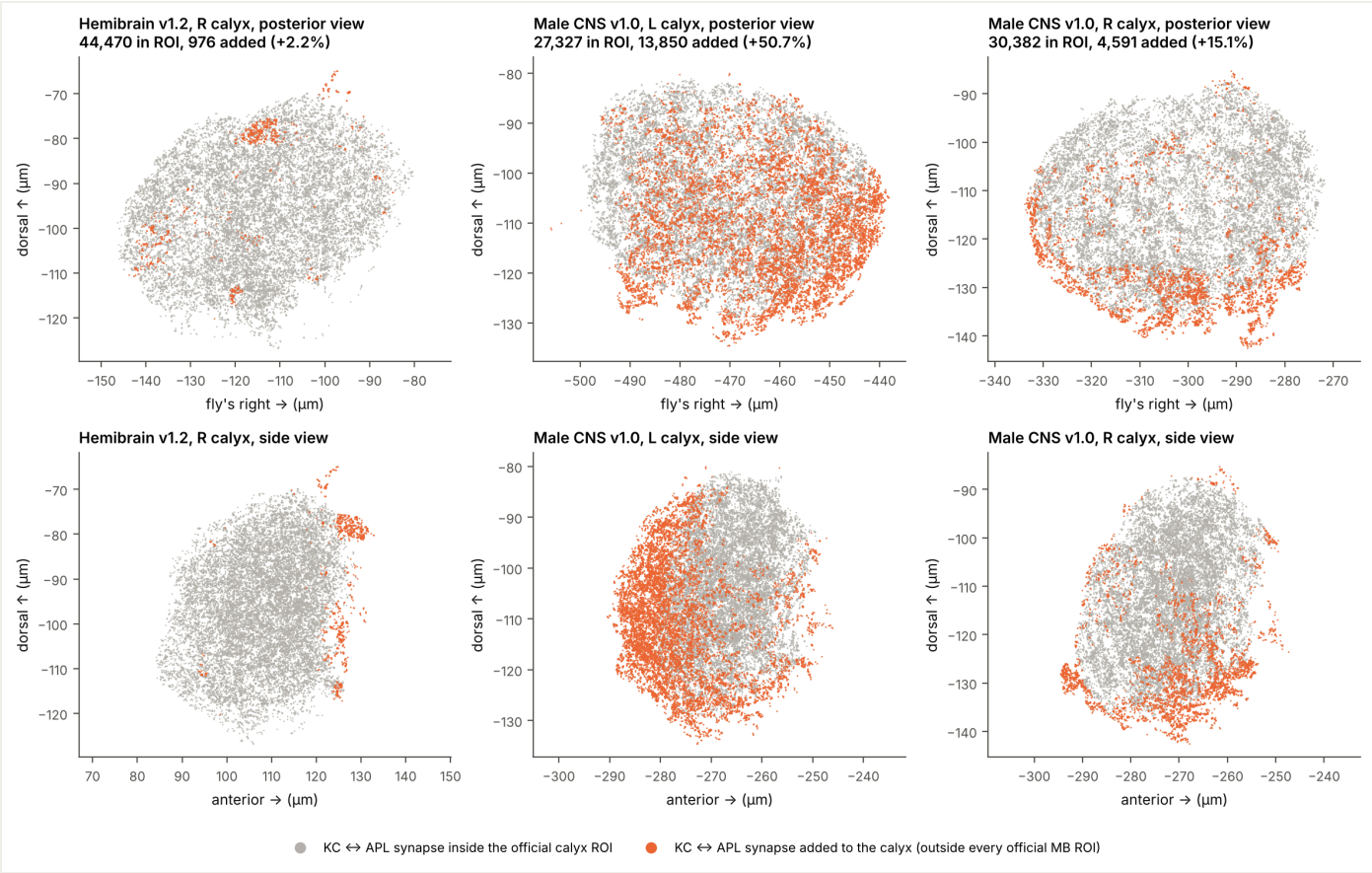


Figure 7. Where the corrected calyx adds KC ↔ APL synapses. Every KC ↔ APL synapse (both directions) in the corrected calyx, at its KC-side site, for the three calyces used by the model. Grey = inside the dataset's official calyx ROI; orange = added by the rule above. Top row: posterior view (axes of Table 4, µm). Bottom row: side view, anterior to the right, dorsal up. Up to 20,000 grey and 12,000 orange points are drawn per panel (random subsample); the counts in the titles are of all KC ↔ APL calyx synapses. In the left male CNS calyx the added synapses form the back (posterior) part of the calyx; in the right one they lie mostly along its ventral and posterior edge.

Caveats

Where an accessory calyx touches the calyx or peduncle, some synapses with γd and αβp KCs within 3 µm of an anchor still join (male CNS 1,291; hemibrain 1,487). They are flagged rows with `partner_type` KCg-d or KCab-p.

The 3 µm steps of the corrected calyx and the 8 µm cut of the claws are choices; the claws were checked against Prat's counts, the corrected calyx by eye on maps.

Male CNS skeletons are coarse (about 0.7 µm between nodes), so distances along them are approximate at that scale.

Where this is used

Report 03 colors KC ↔ APL synapses by the odor response of their claw's glomerulus, using synapses on a claw.

Report 04 counts APL → KC synapses on the claws of each glomerulus (the claw weight q_g).

Reports 05–12 build each KC's input from its claws.

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